Structural Change without an Identified Capital-Composition Elasticity: Chile, 1920–2024

Chapter 2 Empirical Report

July 2026

Abstract

This paper asks whether Chile's capital stocks identify changes in the long-run elasticity of nonresidential structures with respect to machinery and equipment. The canonical annual panel contains 105 observations from 1920 through 2024. The answer is definitive but negative. Neither stable nor break-adjusted cointegration is established, so the data identify no β_{ME} and no period-specific $\beta_{\mathrm{ME},j}$. A separate model of annual accumulation detects a change in 2005, with an analytic 95-percent interval of 1997–2019, but the growth rates do not meet the declared piecewise-stationarity condition. Consequently, the impact and three-year accumulation responses $\eta_{j,0}$ and $\eta_{j,3}$ are also not identified. The capital data therefore reveal historical changes in composition and in the conditional growth relation, but they do not isolate a long-run or short-run elasticity. This distinction prevents persistent stock trends, fitted regression breaks, and descriptive curvature from being mistaken for changes in technique.

1 The answer and its economic meaning

Chile's machinery and equipment stock K^{ME} and its nonresidential structures stock K^{NR} represent different dimensions of accumulation. Machinery is treated here as the intensive margin through which production is reorganized, while structures form the plant envelope within which those techniques operate. This is a maintained analytical restriction, not a claim that the capital-stock data directly measure technology. The empirical question is whether the two observed stocks contain enough independent variation to measure their relationship.

Their log difference defines capital composition:

$$\kappa_t = k_t^{\mathrm{ME}} - k_t^{\mathrm{NR}}. \tag{1}$$

Because this is an identity, the vector $(k_t^{\mathrm{NR}}, k_t^{\mathrm{ME}}, \kappa_t)'$ is exactly singular. The system must therefore be bivariate, $X_t = (k_t^{\mathrm{NR}}, k_t^{\mathrm{ME}})'$, with κ_t used as a derived contrast rather than as a third endogenous variable.

The long-run relation is normalized as

$$k_t^{\mathrm{NR}} = \alpha_j + \beta_{\mathrm{ME},j} k_t^{\mathrm{ME}} + u_t, \qquad t \in (T_{j-1}, T_j]. \tag{2}$$

Within a statistically admissible period j, a 10-percent permanent increase in machinery would correspond to approximately $10\beta_{\mathrm{ME},j}$ percent more structures. A coefficient below one would mean that machinery outpaces structures; a coefficient above one would mean that structures increase more than proportionally. That interpretation requires a stationary long-run error u_t. Without cointegration, a fitted slope between two persistent stocks is not a long-run elasticity.

Table 1 separates three findings that would otherwise be easy to conflate. First, the capital stocks do not establish a stable or period-specific long-run elasticity. Second, the annual growth relation changes around 2005, but its coefficients do not pass the stationarity condition required for an accumulation elasticity. Third, the curvature of κ_t supplies descriptive historical markers only. The paper therefore dates changes in observed capital behavior without assigning an elasticity magnitude that the data cannot sustain.

Table 1: Definitive empirical answer

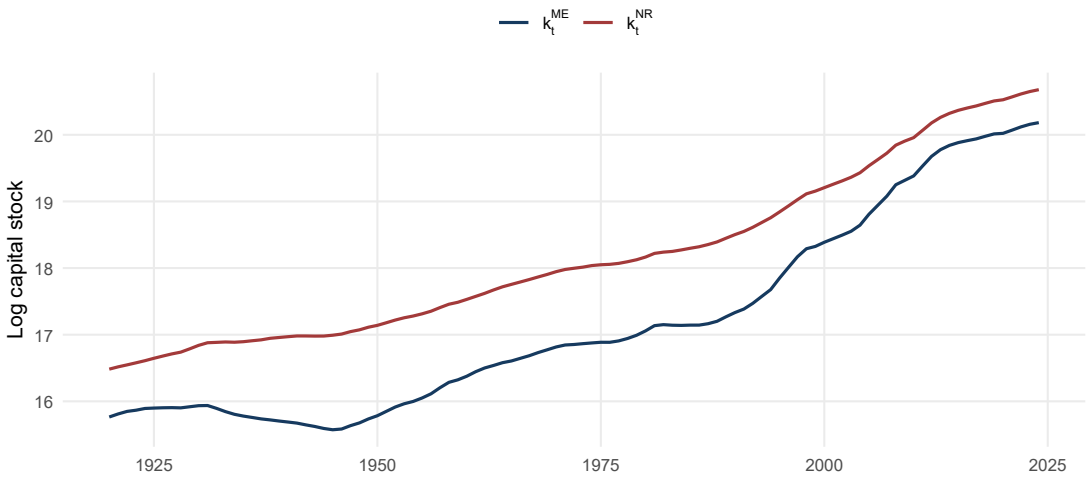
Quantity	Status	Result	Economic meaning
Long-run β_{ME}	Not identified	not identified	the data do not identify a stable long-run structures elasticity with respect to machinery
Changes in $\beta_{\mathrm{ME},j}$	Not identified	not identified	no structural period supports a period-specific cointegrating elasticity
Conditional accumulation η_j	Not identified	not identified	growth-rate persistence prevents an elasticity interpretation
Dynamic-relation break	Diagnostic change	2005 (95% interval 1997–2019)	the conditional growth relation visibly changes, but its coefficients are not admissible elasticities
LOESS curvature dates	Descriptive only	descriptive inflection markers only	changes in the tendency of capital composition, not parametric breaks

Note: A dated change in an observed relation is not automatically an elasticity change. Long-run elasticity requires cointegration and rank one; the conditional accumulation elasticity additionally requires stationary growth rates.

2 Capital stocks and capital composition

The source is the canonical Chile capital panel in constant 2003 CLP. All 105 annual observations from 1920 through 2024 reproduce, and the frozen source hash begins `3a76c95396024d98...`. Older results based on 91 observations or on a 1920–2010 endpoint are not evidence for this sample. Earlier files that labeled ordinary least squares with Newey–West errors or cumulative regressions as FM–OLS or IM–OLS are also excluded. No central result below uses those coefficients.

Figure 1: Chile's machinery and nonresidential structures stocks, 1920–2024

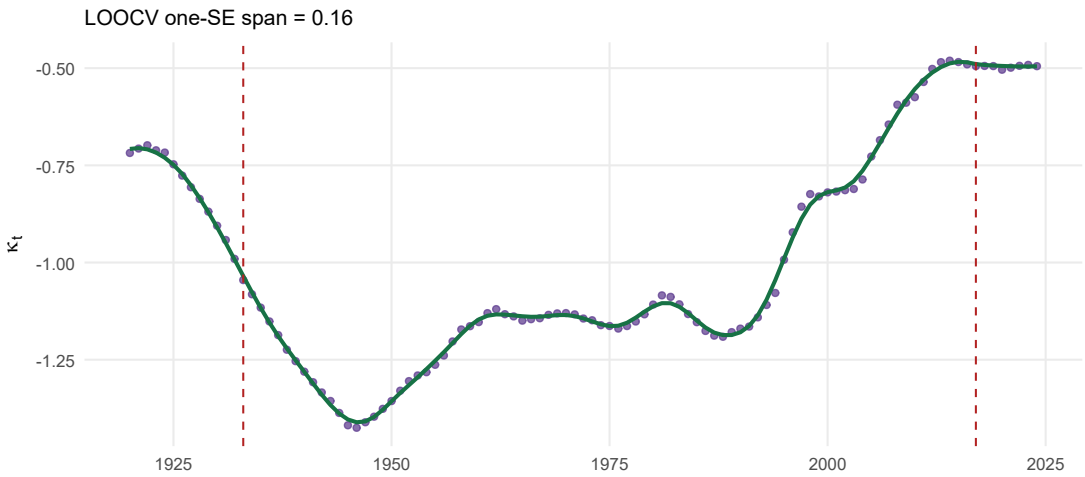


Note: The figure plots the observed log capital stocks before deterministic adjustment. Their common upward movement motivates a long-run test but does not establish cointegration. Both stocks cumulate investment and retirements, so persistence is partly built into their measurement.

The common rise in Figure 1 is economically important but statistically insufficient. Capital stocks are accumulated magnitudes; each observation embeds the preceding investment history. A high correlation can therefore emerge even when the residual combination in equation (2) remains nonstationary. This is why the empirical burden lies first on integration order and cointegration, not on the size of a levels-regression slope.

Figure 2 brings the historical movement down to the observed composition ratio. The scatter shows the data; the line summarizes its local tendency. The second derivative is used only to mark changes in

Figure 2: Observed capital composition and its LOESS tendency



Note: Points are the observed κ_t; the continuous line is a LOESS fit. Zero-crossings of the fit's second derivative occur near 1933, 2017. These dates describe changes in curvature. The fitted line and its derivatives are neither economic time series nor inputs to the parametric break tests.

curvature. It is never unit-root tested, never treated as a stochastic series, and never used to choose the formal coefficient periods. Agreement or disagreement between a curvature date and a parametric date is descriptive comparison, not mutual validation.

3 What the structural-break tests establish

A structural break can concern the path of one stock, the deterministic component of a relation, the slope of a fitted regression, or the cointegrating vector itself. These are different empirical objects. Table 2 states what each procedure can and cannot support.

Table 2: What the structural-break and cointegration procedures establish

Procedure	Economic-statistical question	Permitted claim
Bai–Perron	Where do coefficients in a fitted regression change?	Candidate regression dates; not cointegration
Zivot–Andrews	Does one series reject a unit root after an unknown break?	Integration-order evidence for one series
Gregory–Hansen	Does cointegration appear after one unknown regime change?	One-break cointegration evidence
Kejriwal–Perron	Are coefficients stable in an already cointegrated regression?	Coefficient stability conditional on cointegration
Johansen–Mosconi–Nielsen	What is the rank with known deterministic breaks?	Conditional rank and vector restrictions
L"utkepohl–Saikkonen–Trenkler	What is the rank after one unknown system shift?	Conditional rank with one unknown level shift
Hansen–Johansen	Is an established cointegrating vector recursively constant?	Parameter constancy after rank is established
Phillips–Ouliaris	Does a residual combination reject no cointegration?	Stable cointegration evidence
Lee–Strazicich	Does one series reject a unit root with two admissible breaks?	Break-aware integration-order evidence
`mbreaks`	How are Bai–Perron statistics and intervals computed?	Implementation, not a separate economic test

Note: Each procedure concerns a different statistical object. Evidence about a single series, a fitted regression, a cointegrating residual, and system rank cannot be substituted for one another.

3.1 From an individual series to a long-run relation

Zivot–Andrews asks whether a single series rejects a unit root once one unknown break is admitted under the alternative. In a stylized intercept-and- trend specification,

$$y_t = \mu + \theta DU_t(\lambda) + \beta t + \gamma DT_t(\lambda) + \rho y_{t-1} + \sum_{i=1}^{p} c_i \Delta y_{t-i} + e_t, \tag{3}$$

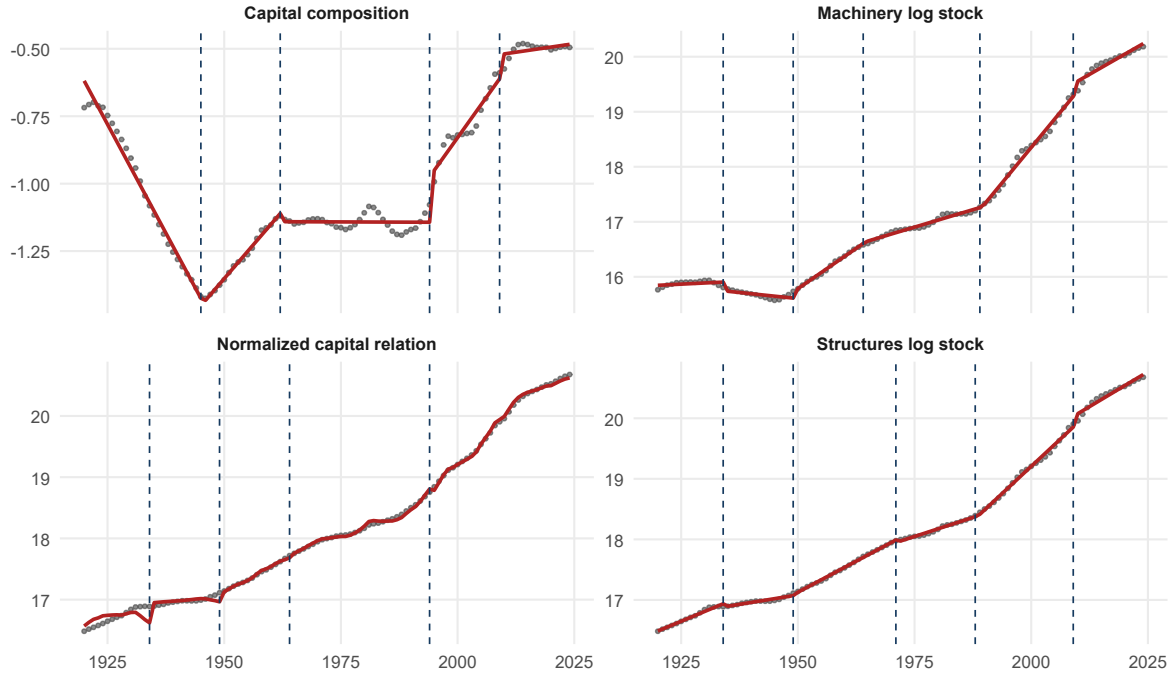
the break fraction λ is selected where the unit-root statistic is most adverse to the null. Rejection informs the integration classification of y_t; it does not show that the slope linking k_t^{NR} to k_t^{ME} changed. Lee–Strazicich provides a complementary two-break LM test that permits breaks under both null and alternative. Neither test, by itself, identifies $\beta_{\mathrm{ME},j}$.

Bai–Perron asks whether coefficients in a stable regression are constant across unknown segments. For m breaks,

$$y_t = x_t'\vartheta + z_t'\delta_j + e_t, \qquad t = T_{j-1}+1, \ldots, T_j, \tag{4}$$

it minimizes the global residual sum of squares subject to a minimum segment length and tests zero against one or more breaks. The procedure is highly useful for locating candidate changes and constructing confidence intervals. In a levels regression with integrated variables, however, ordinary Bai–Perron inference does not establish that the residual is stationary. Its dates cannot, on their own, divide the long-run elasticity.

Figure 3: Bai–Perron segmented fits: candidate changes in the capital relation



Note: Segmented fits and confidence intervals reproduce the candidate evidence for the stock equations, κ_t, and the normalized capital relation. The principal parametric candidate dates are 1945, 1971, 1989. They locate instability in fitted relations; they do not establish cointegration or a change in β_{ME}.

Figure 3 restores the break graph omitted from the earlier report. Its economic value is to show that a constant descriptive relation is implausible over the century. Its inferential limit is equally important: smooth persistent stocks can generate sharply segmented fits even when no stationary long-run error exists.

3.2 From candidate instability to cointegration

Gregory–Hansen tests the null of no cointegration against cointegration with one unknown level, trend, or regime change. A regime-shift version can be written

$$k_t^{\mathrm{NR}} = \alpha_1 + \alpha_2 DU_t(\lambda) + \beta_1 k_t^{\mathrm{ME}} + \beta_2 DU_t(\lambda) k_t^{\mathrm{ME}} + u_t. \tag{5}$$

The statistic takes the most favorable residual-based unit-root value over the trimmed set of possible dates. Rejection would establish one-break cointegration under the chosen deterministic form; the estimated date would then be eligible for system analysis.

Kejriwal–Perron addresses multiple coefficient changes in an already cointegrated regression with integrated regressors. It therefore asks the right stability question only after cointegration is established. The UDmax statistic tests stability against an unknown number of breaks, and sequential supF tests select their count subject to the trimming rule. A large statistic can reject coefficient stability conditional on cointegration; it cannot supply the missing cointegration prerequisite.

Johansen–Mosconi–Nielsen then provides rank inference with known deterministic breaks, while Lütkepohl–Saikkonen–Trenkler treats one unknown system-level shift. Hansen–Johansen recursive statistics concern the constancy of a cointegrating vector after its rank is established. These system procedures answer whether equation (2) belongs to a coherent adjustment system; they are not substitutes for the single-series or residual-cointegration evidence.

4 Long-run cointegration and β_{ME}

The full integration audit applies ADF, Phillips–Perron, KPSS, ERS–GLS, Schmidt–Phillips, and Zivot–Andrews tests to k_t^{NR}, k_t^{ME}, and κ_t in levels, first differences, and second differences, supplemented by break-aware tests. The primary deterministic cases do not produce a common, unambiguous $I(1)$ classification for the two stocks. Conflicting evidence is retained as inconclusive rather than forced into an $I(1)$ designation.

Table 3: Long-run identification evidence

Evidence	Result	Gate	Permitted conclusion
Raw integration order	machinery stock inconclusive; structures stock inconclusive	No	raw capital orders remain inconclusive
Gregory–Hansen	minimum -4.778 against -4.95	No	one-break cointegration is not supported
Phillips–Ouliaris	0 of 6 reject	No	stable cointegration is not supported
No-break Johansen	rank=2	No	the system does not supply an admissible rank-one relation
Independent Kejriwal–Perron	UDmax 50.118 against 12.25	No	stability statistic cannot define coefficient periods
Validated long-run breaks	none	No	no date defines a long-run coefficient period
Stable β_{ME}	not identified	No	not run: integration, cointegration, and rank prerequisites fail
Period $\beta_{\mathrm{ME},j}$	not identified	No	not run: integration, cointegration, and rank prerequisites fail

Note: The independent Kejriwal–Perron statistic exceeds its critical value, but the procedure assumes cointegration. Since neither Gregory–Hansen nor Phillips–Ouliaris establishes cointegration, the stability rejection cannot define coefficient periods.

The residual and system evidence in Table 3 resolves the question. The best Gregory–Hansen statistic is -4.778, which does not cross its relevant 5-percent critical value of -4.95. None of the six Phillips–Ouliaris variants rejects no cointegration. The no-break Johansen system selects rank two rather than the single cointegrating vector required to identify equation (2).

The independent `mbreaks` implementation produces UDmax = 50.118, above the 5-percent critical value 12.25. This corrects the earlier custom computation, which understated the instability evidence. Yet the correction does not create coefficient periods: the Kejriwal–Perron distribution presumes a cointegrated regression, while the preceding tests do not establish one. The statistic therefore says that a stable cointegrating regression would be rejected if such a regression were admissible; it cannot prove that the capital stocks cointegrate.

No date consequently passes the full sequence of integration, break-aware cointegration, rank-one, and residual-dynamics conditions. The actual `cointRegIM` estimator is intentionally not run, because a super-consistent coefficient is not meaningful after those prerequisites fail. FM–OLS and DOLS cannot repair the failure and remain excluded from the central evidence. There is therefore no admissible β_{ME}, no $\beta_{\mathrm{ME},j}$, and no statistical basis for testing whether a reported period coefficient equals one.

The same logic governs κ_t. Global stationarity would require the unit-vector restriction $\beta_{\mathrm{ME}} = 1$, no slope changes, and no permanent deterministic shift in κ_t. Piecewise stationarity would permit level shifts but still require every regime-specific machinery coefficient to equal one. Since no cointegrating vector is identified, these restrictions cannot be evaluated as long-run restrictions. Visual proximity of a fitted coefficient to one would not establish stationarity.

5 Stationary accumulation and η_j

Long-run non-identification does not rule out a separate relation between annual accumulation rates. The relevant stationary model is

$$\Delta k_t^{\mathrm{NR}} = c_j + \sum_{\ell=1}^{p} a_\ell \Delta k_{t-\ell}^{\mathrm{NR}} + \sum_{\ell=0}^{p} b_{j\ell} \Delta k_{t-\ell}^{\mathrm{ME}} + e_t. \tag{6}$$

Here $\eta_{j,0} = b_{j0}$ would measure the conditional impact response of structures growth to machinery growth, while $\eta_{j,3}$ would cumulate the dynamic response through three years. These are conditional accumulation elasticities, not the long-run cointegrating coefficient in equation (2).

The lag order is chosen on a common sample. BIC selects $p = 1$; the resulting equation passes the four-lag residual serial-correlation test with $p = 0.504$. Allowing the intercept and machinery-response coefficients to change, the break test gives UDmax = 34.064, above its 5-percent critical value 14.23. The sequential procedure rejects zero against one break but not one against two, locating the change in 2005 with a 95-percent interval of 1997–2019.

Table 4: Break tests for the dynamic accumulation relation

Test	Statistic	5% critical	Reject	Sequential implication
UDmax	34.064	14.23	Yes	2005 (1997–2019)
supF(1|0)	34.064	13.98	Yes	continue
supF(2|1)	9.503	15.72	No	stop
supF(3|2)	15.324	16.83	No	stop

Note: The dynamic equation uses one annual lag, selected by BIC on a common sample, and passes a four-lag residual serial-correlation test. These statistics locate a visible conditional change. They do not become elasticity evidence because the growth-rate stationarity condition fails.

This is strong evidence that the conditional growth relation is not constant. It is not yet an elasticity result. The statistical meaning of the coefficients in equation (6) depends on stationary growth rates within the deterministic specification used to define the break. The complete battery is therefore repeated after allowing a 2005 mean shift.

For Δk_t^{NR}, three of the six primary test families support stationarity; ERS is conflicting, while Schmidt–Phillips and Zivot–Andrews retain a unit root. For Δk_t^{ME}, only ADF and KPSS support stationarity; Phillips–Perron conflicts and the remaining families retain a unit root. The declared requirement of four concordant families is not met for either series, and the two-break Lee–Strazicich

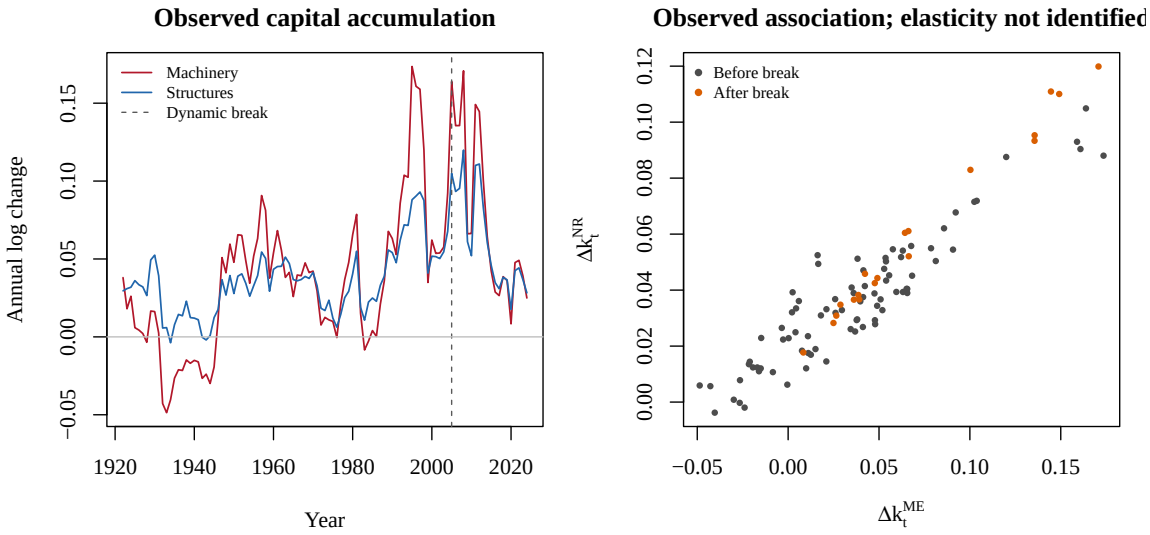
Table 5: Stationarity evidence for the capital-growth rates after the 2005 mean shift

Series	ADF	PP	KPSS	ERS	SP	ZA	Conclusion
Δk^{NR}	stationary	stationary	stationary	conflict	unit root	unit root	Inconclusive
Δk^{ME}	stationary	conflict	stationary	unit root	unit root	unit root	Inconclusive

Note: A family verdict summarizes its declared primary deterministic cases so that densely parameterized test families receive no extra weight. Four concordant families are required. The disagreement therefore remains substantive rather than being resolved by majority choice among individual configurations.

tests do not reverse that verdict. The evidence remains inconclusive instead of being resolved by counting the many specifications inside one test family.

Figure 4: Annual capital accumulation and the 2005 diagnostic break



Note: The left panel plots observed annual log changes; the right panel displays their contemporaneous association. The vertical line marks the diagnostic 2005 change. No fitted response line is shown because the stationarity condition needed to interpret its slope as η_j is not satisfied.

Table 6: Can the conditional accumulation elasticity be reported?

Span	N	$\eta_{j,0}$	$\eta_{j,3}$	Adjacent test	Empirical verdict
1922-2005	84	–	–	–	Not identified: growth-rate stationarity is inconclusive
2006-2024	19	–	–	–	Not identified: growth-rate stationarity is inconclusive

Note: $\eta_{j,0}$ would be the impact response and $\eta_{j,3}$ the cumulative three-year response. Both are intentionally blank because the growth rates do not pass the declared piecewise-stationarity rule. The 2005 break remains a feature of the observed dynamic relation, not an estimated elasticity change.

The blank coefficient cells in Table 6 are empirical results, not missing calculations. Estimating separate period regressions would manufacture magnitudes after the governing condition failed, especially in the 19-observation post-2005 period. The conditional bivariate VAR and Granger-exclusion tests are also not reported because they require the same stationary system. Resampling cannot remedy this problem: bootstrap inference can approximate the distribution of an admissible statistic, but it cannot turn persistent growth rates into stationary variables. No bootstrap is therefore necessary or appropriate for the reported verdict.

6 Historical interpretation

The evidence establishes structural change in Chilean capital accumulation without identifying its elasticity. The capital stocks visibly change their relative tendency over the century, ordinary segmented regressions reject constancy, and the dynamic growth relation changes around 2005. These findings reject a timeless descriptive mapping from machinery into structures. They do not reveal by how much a permanent machinery expansion changes structures in the long run, nor do they supply a stationary short-run response.

This boundary matters for the choice-of-technique interpretation. A rising κ_t is consistent with machinery deepening relative to the plant envelope, but it does not by itself identify the behavioral mechanism. Capital-stock construction combines investment, retirement, relative prices, and inherited stocks. With only the two stock series, secular measurement persistence and economic adjustment cannot be separated tightly enough to assign a coefficient magnitude. The maintained intensive/extensive distinction supplies the economic question; it cannot supply the missing statistical variation.

The 2005 change has a similarly bounded interpretation. It indicates that the observed conditional association between the two accumulation rates changes within a broad 1997–2019 interval. The interval is too wide for a single institutional date to be inferred from the capital data alone, and the nonstationary evidence blocks a parameter interpretation. The result is historically suggestive but neither causal nor an estimate of technique.

7 Conclusion

The canonical 1920–2024 Chile capital panel does not identify the order of magnitude of β_{ME}, either as one stable coefficient or as period-specific coefficients across structural breaks. The decisive reason is not the absence of visible breaks. On the contrary, instability is strong: the independent Kejriwal–Perron statistic and the segmented fits reject constancy. The reason is that stable and one-break cointegration are not established and the system does not supply an admissible rank-one vector. Candidate dates therefore cannot become long-run coefficient periods.

The separate accumulation analysis reaches the other predetermined outcome. A single dynamic change is detected in 2005, but neither capital-growth rate passes the declared piecewise-stationarity condition. The evidence therefore does not identify $\eta_{j,0}$ or $\eta_{j,3}$. Chile's capital data establish historical change in composition and accumulation, but they identify neither a long-run elasticity nor a conditional accumulation elasticity. That non-identification is the definitive answer supplied by the available capital stocks.

A Compact methodological record

A.1 Integration and break adjudication

The unit-root audit records deterministic case, lag, statistic, and critical value for ADF, Phillips–Perron, KPSS, ERS–GLS, Schmidt–Phillips, and Zivot–Andrews tests in levels, first differences, and second differences. Test families receive one primary verdict per deterministic case so that families with more tuning choices do not dominate. Break-aware Lee–Strazicich results supplement, rather than replace, this evidence. A deterministic step may absorb a level shift; it is not presumed to convert a genuine $I(2)$ process into $I(1)$.

The parametric evidence uses a 15-percent minimum segment and no more than three breaks. Ordinary Bai–Perron dates are candidate changes. Gregory–Hansen supplies one-break residual cointegration evidence. Kejriwal–Perron supplies multiple-break stability evidence conditional on cointegration. Known deterministic shifts belong in the Johansen–Mosconi–Nielsen system, while an unknown system shift belongs in the Lütkepohl–Saikkonen–Trenkler procedure. No LOESS curvature date enters these computations.

A.2 Coefficient construction and inference

Had validated long-run breaks existed, period coefficients would have been estimated jointly with initial-break-state-centered interactions:

$$k_t^{\mathrm{NR}} = \alpha_1 + \beta_1 k_t^{\mathrm{ME}} + \sum_{b=1}^{m} \gamma_b D_{b,t} + \sum_{b=1}^{m} \theta_b D_{b,t}\big(k_t^{\mathrm{ME}} - k_{T_b}^{\mathrm{ME}}\big) + u_t, \tag{7}$$

where $\beta_{\mathrm{ME},1} = \beta_1$ and $\beta_{\mathrm{ME},j} = \beta_1 + \sum_{b<j} \theta_b$. Here γ_b denotes deterministic step controls in the single equation; the system form uses $\mathbf{\Gamma D}_t$. Uncentered and sample-mean-centered versions would be numerical comparators assessed by confidence-band overlap and condition numbers. Residualization is not used. Since the cointegration conditions fail, these coefficients and their cumulative covariance are not estimated.

Analytic break-aware and HAC inference is the default. Restricted-null residual bootstrap inference would be reserved for an eligible estimated-break system lacking analytic rank inference, material break-selection uncertainty not covered analytically, or an indispensable period with at most 30 observations. None of those situations arises after the identification conditions fail, so the number of bootstrap replications used is zero.

A.3 System interpretation

The bivariate error-correction representation underlying the system checks is

$$\Delta X_t = \mathbf{\Pi} X_{t-1} + \sum_{i=1}^{p-1} \mathbf{\Gamma}_i \Delta X_{t-i} + \mathbf{\Gamma D}_t + \varepsilon_t, \qquad \mathbf{\Pi} = \boldsymbol{\alpha}\boldsymbol{\beta}'. \tag{8}$$

For two persistent capital stocks, rank zero means that no stationary long-run combination enters the system. Rank one is the economically relevant case: $\boldsymbol{\beta}$ contains the normalized capital-composition relation and $\boldsymbol{\alpha}$ records how each stock responds to its disequilibrium. Rank two makes the levels system full rank and therefore does not deliver the single common-trend relation required to interpret β_{ME}. The no-break rank-two result is consequently not evidence of a stronger form of the desired cointegration; it shows that the proposed $I(1)$, rank-one representation is not supported.

Known step shifts enter through $\mathbf{\Gamma D}_t$, while slope changes belong in the cointegrating vector and require break-aware rank inference. A deterministic shift can alter the mean of κ_t, but it cannot be used mechanically to remove stochastic persistence. Adjustment coefficients, weak-exogeneity tests, recursive constancy, and companion-root diagnostics become interpretable only after an admissible rank is established. They are therefore not reported as isolated system statistics after the rank-one condition fails.

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